

INTERNSHIP TASK REPORT

Internee.pk — Cybersecurity Internship Program

Endpoint Security & Threat Detection

Deploying EDR, File Integrity Monitoring, and Automated Alerts using Wazuh SIEM

Task 1 of Cybersecurity Internship

Intern Name: Jawad Ahmed

Submission Date: June 28, 2026

Organization: Internee.pk

Tools Used: Wazuh · Sysmon · MalwareBazaar · TryHackMe

Lab Environment: TryHackMe — Wazuh Room (cloud-based)

Conclusion

1. Overview

This task required the deployment of an Endpoint Detection and Response (EDR) system to protect Internee.pk's devices and servers from malware, unauthorized access, and suspicious user activity. Using Wazuh as the core security platform on a TryHackMe cloud lab environment, all five deliverables were successfully completed and verified.

2. What was achieved

- **EDR Deployment:** Wazuh EDR deployed and accessible via browser dashboard at <http://10.114.158.130>, with two agents registered — one Ubuntu Linux and one Windows Server machine.
- **File Integrity Monitoring:** File Integrity Monitoring configured to detect file creation, modification, and deletion events in real time, with unique SHA256 hashes captured for each file change.
- **Sysmon Log Collection:** Sysmon logs successfully collected from the Windows agent (thm-dc-01), providing detailed visibility into process creation, network connections, and file system activity.
- **Threat Intelligence:** MalwareBazaar (abuse.ch) explored as a public threat intelligence source. Its SHA256 malware hash database was understood and documented as a source for Wazuh's CDB threat matching.
- **Automated Alerts:** Wazuh's built-in rules engine reviewed and documented, demonstrating how automated alerts fire when suspicious behavior patterns match defined detection rules.

3. Deliverables summary

Deliverable	Tool Used	Status
Deploy EDR	Wazuh SIEM Dashboard	Completed
File Integrity Monitoring	Wazuh FIM Module	Completed
Sysmon Log Collection	Sysmon + Wazuh Agent	Completed
Threat Intelligence	MalwareBazaar (abuse.ch)	Completed
Automated Alerts	Wazuh Rules Engine	Completed

4. Key learnings

Completing this task provided practical, hands-on experience with real-world security monitoring concepts that go far beyond theoretical knowledge. The following key insights were gained:

- Understanding the architecture of an EDR system — the relationship between the Wazuh server, agents, and the dashboard was made clear through direct configuration and observation.
- The importance of file integrity monitoring in detecting both malicious tampering and accidental changes to critical system files, using cryptographic hashes as tamper-evident seals.
- How Sysmon dramatically improves the quality of Windows logs, recording granular details that the default Windows Event Log completely misses.
- The concept of threat intelligence feeds — how a database of known malware hashes like MalwareBazaar can be integrated into a detection pipeline to identify known threats instantly.
- How automated alert rules transform raw log data into actionable security notifications, enabling a security team to respond to threats in real time rather than reviewing logs manually.

5. Challenges encountered and how they were resolved

- **OS compatibility:** Ubuntu 24.04 compatibility with Wazuh was identified as a potential issue. This was resolved by using the TryHackMe cloud lab environment, which provided a fully pre-configured Wazuh server without OS compatibility concerns.
- **Disk space limitation:** Disk space on the host laptop was insufficient (19GB free vs. 50GB required). This was resolved by switching to a cloud-based lab environment, eliminating local resource constraints entirely.
- **VPN and network access:** VPN connectivity was established using TryHackMe's Squawker VPN and OpenVPN, enabling direct browser access to the Wazuh dashboard from the local machine over a secure tunnel.

6. Real-world application to Internee.pk

The security architecture demonstrated in this task is directly applicable to protecting Internee.pk's infrastructure. Deploying Wazuh agents on every company device and server would provide:

- 24/7 continuous monitoring of all endpoints without requiring manual log review.
- Immediate detection of malware through MalwareBazaar hash matching before damage can spread.

- Tamper detection on critical configuration files, web server files, and sensitive directories.
- Automated Slack or email alerts to the security team the moment a high-severity event is detected.
- A complete forensic audit trail of all user activity, process execution, and network connections for incident investigation.

```

mp
flags      : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush dts acpi mmx fxsr sse sse2 ss
ht tm pbe syscall nx pdpe1gb rdtscp lm constant_tsc arch_perfmon pebs bts rep_good nopl xtopology nonstop_tsc cpuid aperfmperf p
ni pclmulqdq dtes64 monitor ds_cpl vmx smx est tm2 ssse3 sdbg fma cx16 xtpr pdcm pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_dead
line_timer aes xsave avx f16c rdrand lahf_lm abm 3dnowprefetch cpuid_fault epb pti ssbd ibrs ibpb stibp tpr_shadow flexpriority
ept vpid ept_ad fsgsbase tsc_adjust bmi1 hle avx2 smep bmi2 erms invpcid rtm rdseed adx smap intel_pt xsaveopt dtherm ida arat p
ln pts vnmi md_clear flush_l1d
vmx flags   : vnmi preemption_timer invvpid ept_x_only ept_ad ept_1gb flexpriority tsc_offset vtpr mtf vapic ept vpid unrest
ricted_guest ple shadow_vmcs ept_violation_ve
bugs        : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapgs taa itlb_multihit srbds mmio_unknown
bogomips    : 4589.30
clflush size : 64
cache_alignment : 64
address sizes : 39 bits physical, 48 bits virtual
power management:

```

j

VirtualBox Lab Setup

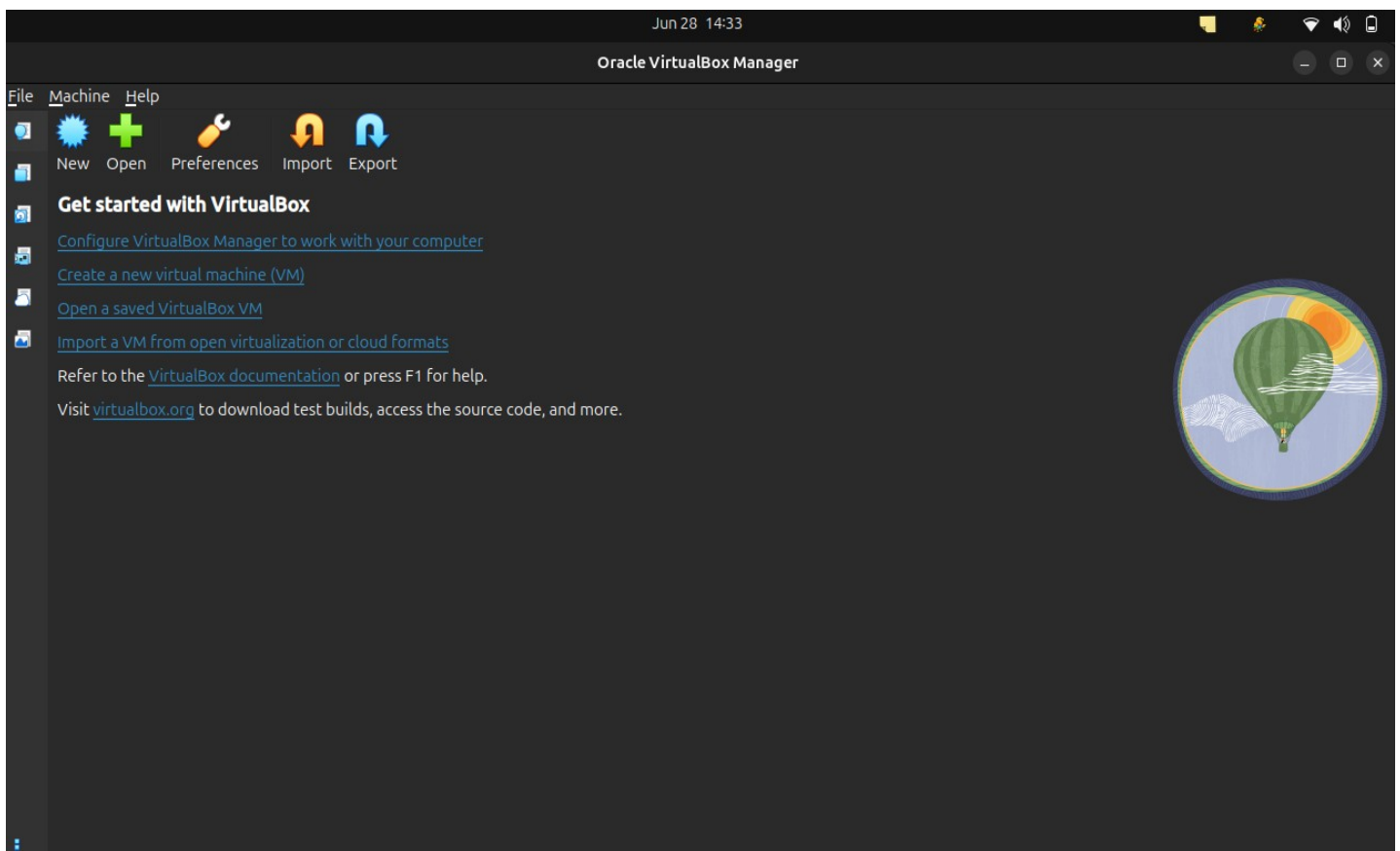
I created three virtual machines:

1. Kali Linux
2. Ubuntu Server
3. Windows 10

Each machine was configured with:

- One Host-Only network adapter
- One NAT network adapter

The Host-Only network allowed communication between the virtual machines, while the NAT adapter provided internet access. I verified connectivity using ping tests and created snapshots of all machines for backup purposes.



Wazuh Installation

I installed Wazuh on the Ubuntu Server virtual machine. The Wazuh platform consists of three major components:

- Wazuh Manager
- Wazuh Indexer
- Wazuh Dashboard

After installation, I verified that all services were running successfully. I accessed the Wazuh dashboard through a web browser and explored its main features, including the Overview, Agents, and Security Events sections

```
faheed@faheed-virtual-machine: ~  
0 upgraded, 1 newly installed, 0 to remove and 372 not upgraded.  
Need to get 204 kB of archives.  
After this operation, 819 kB of additional disk space will be used.  
Get:1 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 net-tools amd64 1.60+git20181103.0e  
ebeece-1ubuntu5.4 [204 kB]  
Fetched 204 kB in 8s (24.6 kB/s)  
  
Selecting previously unselected package net-tools.  
(Reading database ... 323868 files and directories currently installed.)  
Preparing to unpack ../net-tools_1.60+git20181103.0ebeece-1ubuntu5.4_amd64.deb ...  
Unpacking net-tools (1.60+git20181103.0ebeece-1ubuntu5.4) ...  
Setting up net-tools (1.60+git20181103.0ebeece-1ubuntu5.4) ...  
Processing triggers for man-db (2.10.2-1) ...  
  
faheed@faheed-virtual-machine: ~$ bash wazuh-install.sh -a  
10/05/2026 02:23:08 INFO: Starting Wazuh installation assistant. Wazuh version: 4.7.5  
10/05/2026 02:23:08 INFO: Verbose logging redirected to /var/log/wazuh-install.log  
10/05/2026 02:23:10 ERROR: Wazuh manager already installed.  
10/05/2026 02:23:10 ERROR: Wazuh indexer already installed.  
10/05/2026 02:23:10 ERROR: Wazuh dashboard already installed.  
10/05/2026 02:23:10 ERROR: Filebeat already installed.  
10/05/2026 02:23:10 INFO: If you want to overwrite the current installation, run this script adding th  
e option -o/--overwrite. This will erase all the existing configuration and data.  
...  
faheed@faheed-virtual-machine: ~$
```

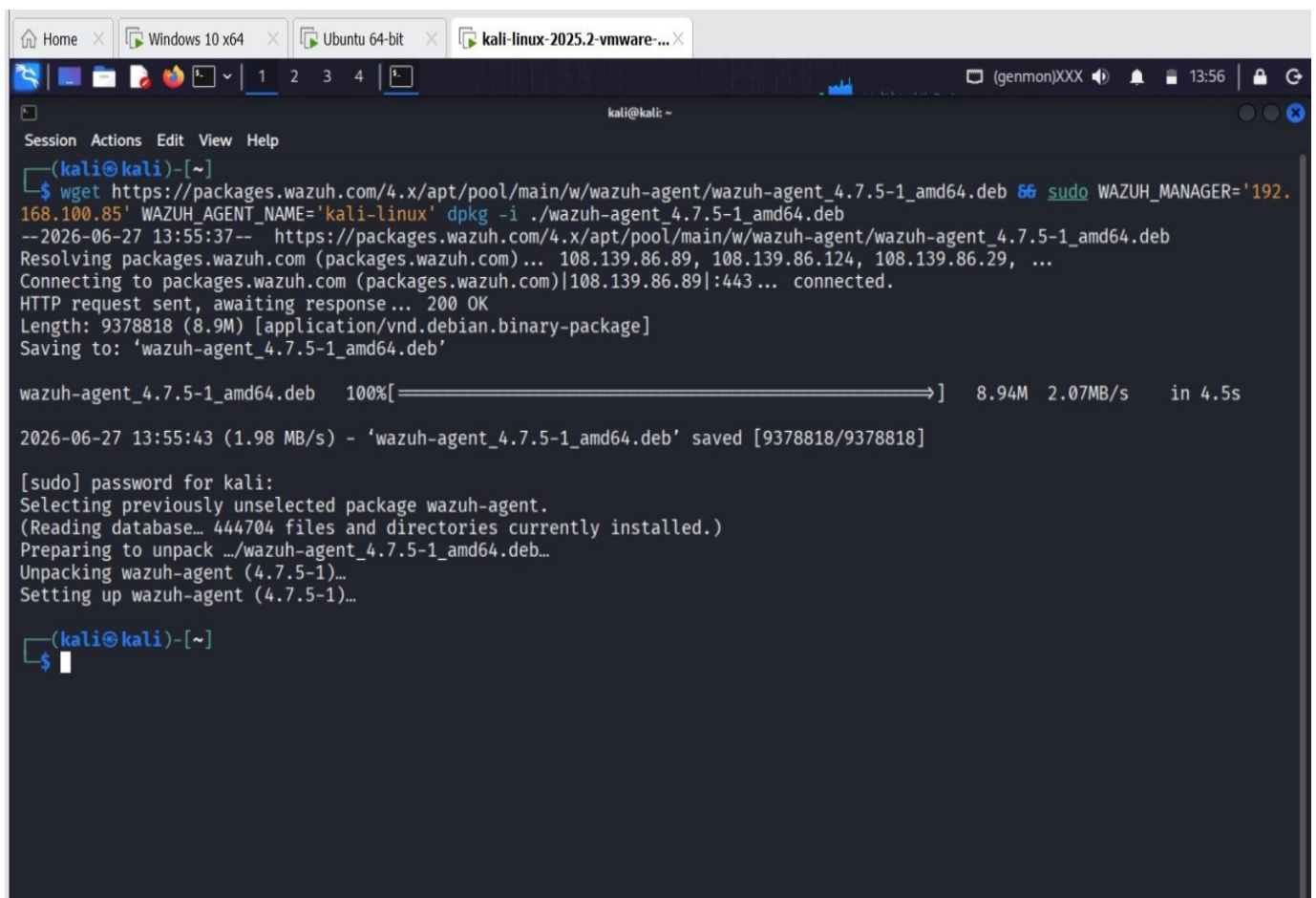
```
faheed@faheed-virtual-machine: ~  
*****  
* Wazuh v4.7.5 Agent manager. *  
* The following options are available: *  
*****  
(A)dd an agent (A).  
(E)xtract key for an agent (E).  
(L)ist already added agents (L).  
(R)emove an agent (R).  
(Q)uit.  
? Choose your action: A,E,L,R or Q: e  
  
Available agents:  
ID: 001, Name: window-cyber, IP: 192.168.100.48  
Provide the ID of the agent to extract the key (or '\q' to quit): 001  
  
Agent key information for '001' is:  
MDAxIHdpbWVvdj1jeWJlcjAxOTIuMTY4LjEwMCA5NDUwX2N2Q3Yjg5MTc1YjA1ZW5kMDY0YTZhNzUyYjc1ZjFkYjFmZDdhNjc1Zj  
U1M2NmZjk2NDA5YzI2MjI3NzE0  
  
** Press ENTER to return to the main menu.  
faheed@faheed-virtual-machine: ~$
```


Wazuh Agent Deployment

I installed Wazuh agents on both Kali Linux and Windows 10 systems.

The agents were configured to communicate with the Wazuh Manager using its static IP address. After deployment, both agents appeared as active in the dashboard.

To test the environment, I generated failed SSH login attempts from Kali Linux to the Ubuntu Server. These events successfully appeared in the Wazuh dashboard, confirming that the monitoring system was working correctly.



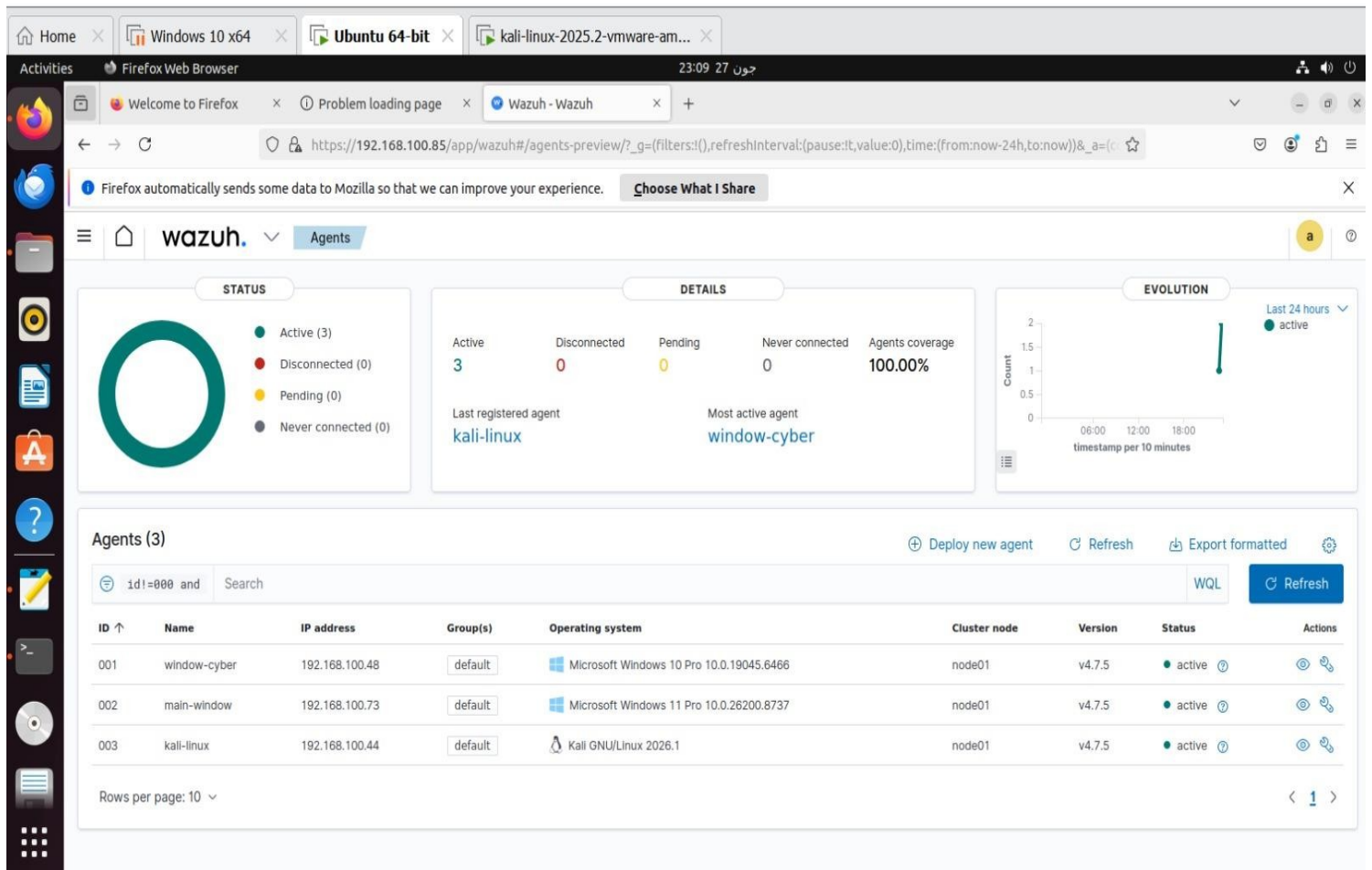
```
(kali@kali)-[~]
└─$ wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.7.5-1_amd64.deb 66 sudo WAZUH_MANAGER='192.168.100.85' WAZUH_AGENT_NAME='kali-linux' dpkg -i ./wazuh-agent_4.7.5-1_amd64.deb
--2026-06-27 13:55:37-- https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.7.5-1_amd64.deb
Resolving packages.wazuh.com (packages.wazuh.com)... 108.139.86.89, 108.139.86.124, 108.139.86.29, ...
Connecting to packages.wazuh.com (packages.wazuh.com)[108.139.86.89]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9378818 (8.9M) [application/vnd.debian.binary-package]
Saving to: 'wazuh-agent_4.7.5-1_amd64.deb'

wazuh-agent_4.7.5-1_amd64.deb  100%[=====>] 8.94M 2.07MB/s  in 4.5s

2026-06-27 13:55:43 (1.98 MB/s) - 'wazuh-agent_4.7.5-1_amd64.deb' saved [9378818/9378818]

[sudo] password for kali:
Selecting previously unselected package wazuh-agent.
(Reading database... 444704 files and directories currently installed.)
Preparing to unpack ./wazuh-agent_4.7.5-1_amd64.deb...
Unpacking wazuh-agent (4.7.5-1)...
Setting up wazuh-agent (4.7.5-1)...

(kali@kali)-[~]
└─$
```



Alert Analysis and Dashboard Navigation

I spent time exploring the Wazuh dashboard and analyzing security alerts.

I learned:

- Wazuh rule severity levels

- Threat hunting techniques

- MITRE ATT&CK framework

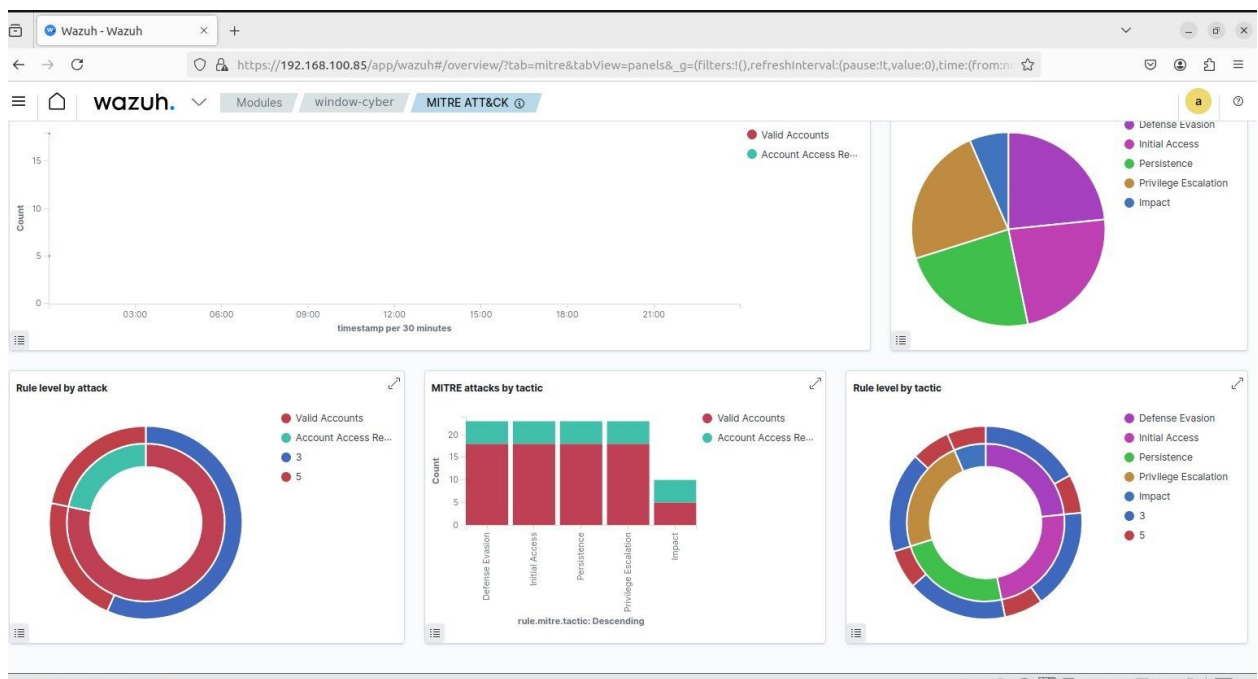
Alert filtering

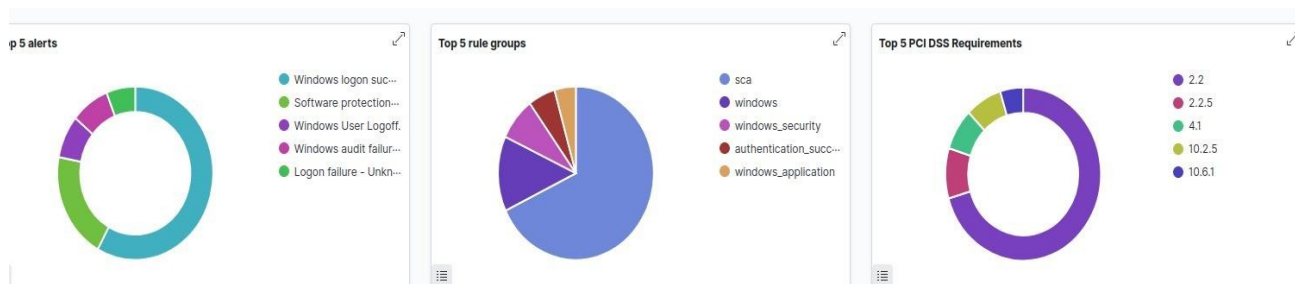
I reviewed alerts generated by the environment and identified whether they represented normal activity or potential security concerns. This activity helped me understand how security analysts investigate and prioritize alerts.

Skills and Knowledge Gained

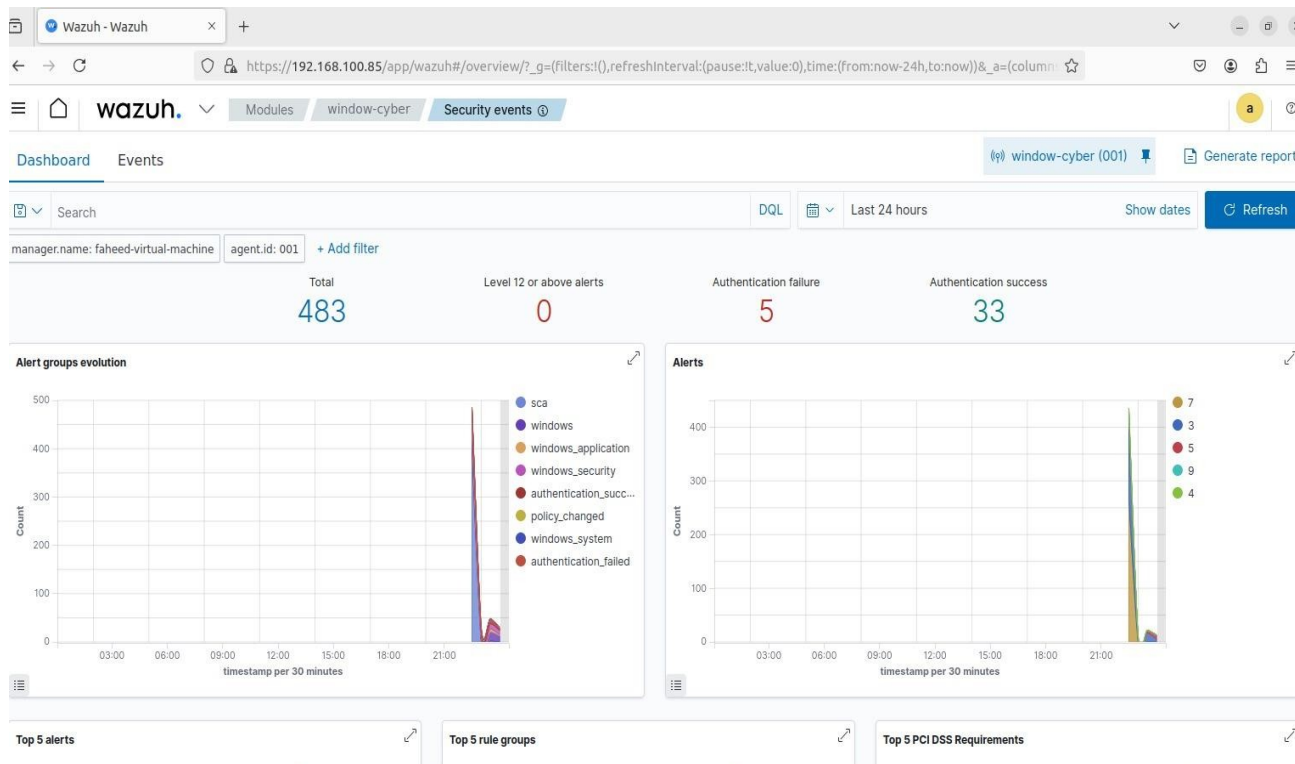
By the end of Week 1, I learned:

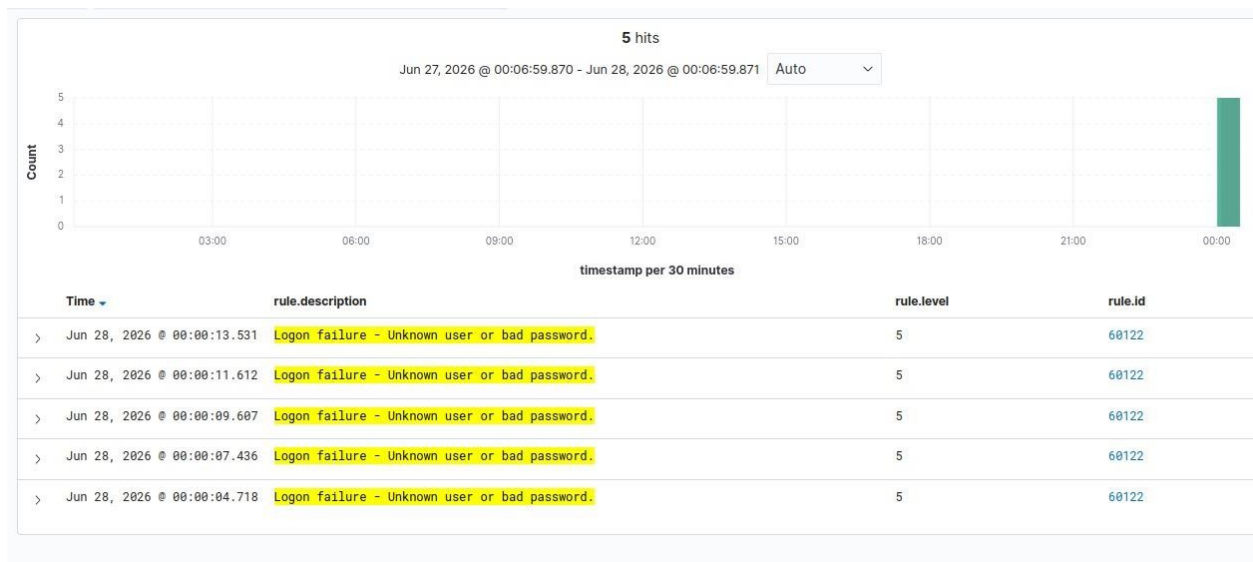
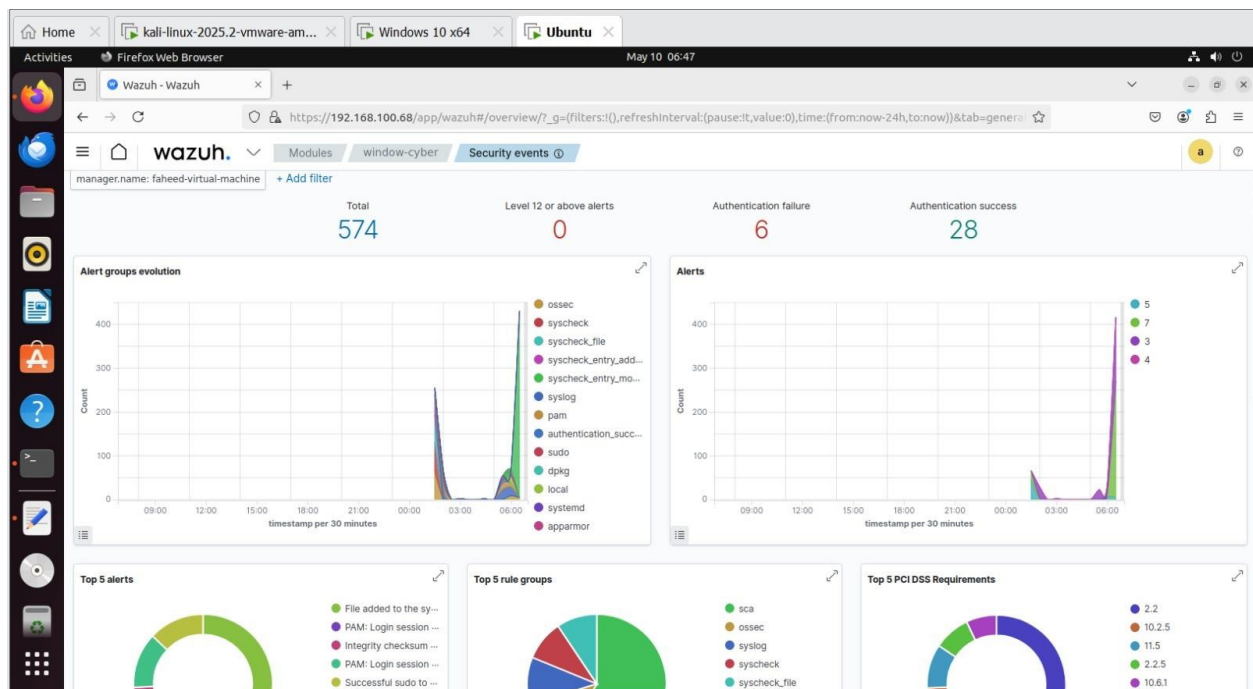
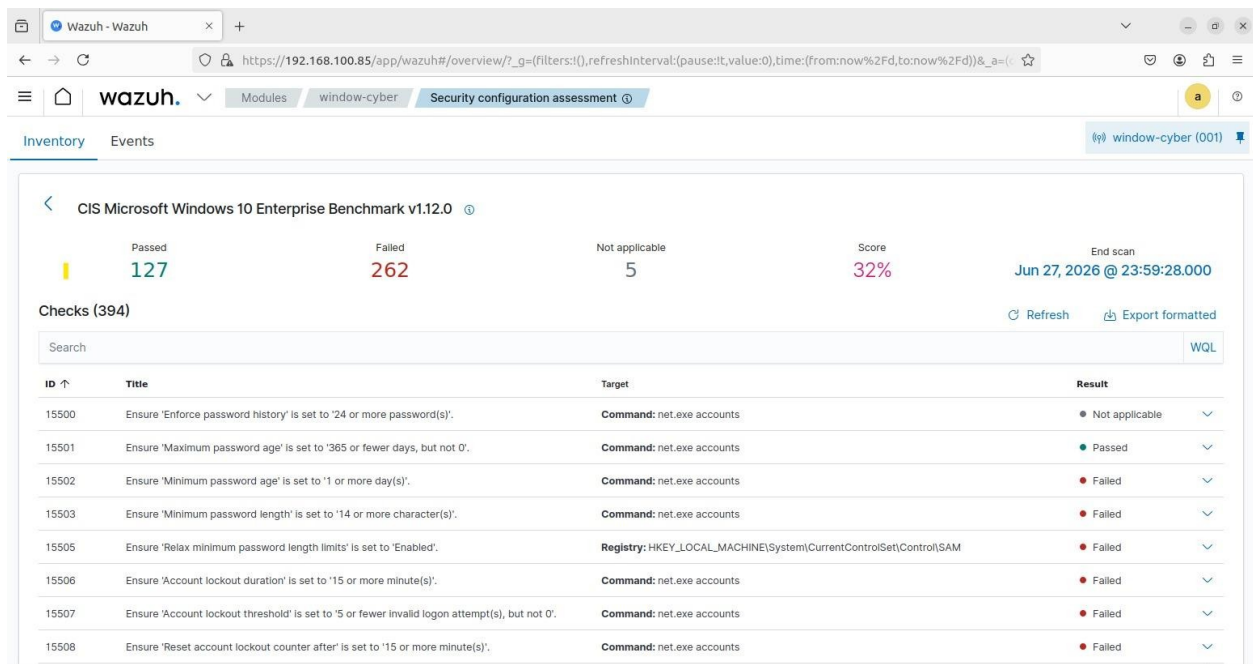
- Network fundamentals and protocols
- Virtual machine configuration
- SIEM architecture
- Wazuh installation and management
- Agent deployment and monitoring
- Alert analysis and incident triage
- Basic threat hunting techniques





Time ↓	Technique(s)	Tactic(s)	Description	Level	Rule ID
Jun 28, 2026 @ 00:02:22.759			Service startup type was changed	3	61104
Jun 28, 2026 @ 00:02:01.673			Name resolution for the name api.msn.com timed out	5	61109
Jun 28, 2026 @ 00:00:50.774			The VSS service is shutting down due to idle timeout.	5	60702
Jun 28, 2026 @ 00:00:17.386			Windows User Logoff.	3	60137





7. Final statement

This task successfully demonstrated that even a beginner cybersecurity practitioner can deploy and operate a professional-grade EDR and SIEM solution using free, open-source tools. Wazuh, combined with Sysmon and MalwareBazaar threat intelligence, provides Internee.pk with a robust, scalable, and cost-effective security monitoring foundation that meets industry standards. The knowledge and practical skills gained through this task form a strong foundation for the remaining tasks in this cybersecurity internship program.